

MISSION POSSIBLE

DESCRIPTION: Teams will design, build, test and bring a Rube Goldberg-like device which incorporates up to 12 unique Action Transfers and uses five forms of energy in accomplishing a given task in 60 seconds. Devices will be constructed prior to competition. This is a **PRE-BUILT EVENT**.

NUMBER OF PARTICIPANTS: Maximum of 2

IMPOUND: Yes

COMPETITORS BRING:

- Device
- 2 copies of the Action Sequence List (ASL)
- Safety goggles
- Any materials needed to repair device

THE COMPETITION:

CONSTRUCTION:

1. GENERAL REQUIREMENTS:

- The device must fit inside an imaginary box **75 cm³** and must fit through a single door opening. Devices not meeting these requirements will be placed in Tier 2. Device may be on wheels, but must still comply with size requirements
- The device may use ambient room light and must not depend on direct sunlight to operate. All other sources of energy and actions must take place within the imaginary box before, during, and after the device's operation.
- All parts, pieces, substances must remain within the device at all times. Any violation will result in a 20 point penalty per offense.**
- Students must be able to move the device from one location to another (from impound to testing).
- For measured actions, team must have a measurement tool to show the required distance (this can be a portion of a paper centimeter ruler)

2. SCORABLE ACTIONS:

- Start Action: **(40 Points)** - The device will be started by dropping a golf ball from completely above the device to initiate a subsequent action. This should be clearly identified on the ASL for points.
- Scorable Actions: **(20 points each for a total of 240)** - each scorable action needs to initiate the next scorable action.
 - Use a 3:1 gear system to initiate an action
 - Use a wheel and axle to raise an object at least 10 cm vertically, that object should then initiate an action **(the measurement should be marked on the device)**
 - Transfer gravitational potential energy to kinetic energy to initiate an action
 - Use chemical energy to initiate an action
 - Knock over a series of at least 5 free-standing, commercial dominoes, the last of which initiates an action
 - Use a pulley system to raise an object at least 10 cm vertically, that object should then initiate an action **(the measurement should be marked on the device)**
 - Push or pull an object up an inclined plane to initiate an action
 - Use wind energy to initiate an action
 - Use water to raise an object at least 5 cm to initiate an action (the measurement should be marked on the device)**
 - Pull a wedge from underneath an object to initiate an action
 - Blow up a balloon and use that to initiate an action

- xii. Use a wheel and axle to move an object at least 10 cm horizontally, that object should then initiate an action **(the measurement should be marked on the device)**
- xiii. *Actions must be designed to contribute to the final action and should be consecutive. Actions that are parallel or dead-end actions will not count for points.*
- xiv. *Scorable actions cannot be combined and should be identified on the ASL. For example, if you use a funnel to transfer gravitational energy to kinetic energy, the team must identify which of those is the single scorable action. A battery can not count as both chemical and electrical energy if initiating the same action.*
- xv. *Energy sources (i.e. springs, mousetraps) must be set prior to starting the device in the 5 minute maximum set up time.*
- xvi. *Heat energy may be used but must not have a flame.*
- xvii. *No electric device may have or use an electric potential difference of more than 9 volts.*
- xviii. *If electric devices used for heat are rechargeable, documentation must be submitted to confirm output is no more than 9 volts.*
- c. Final Action: **(40 points)** Teams will design the device to complete the final task of **raising a flag completely above the top of the device**, at the end of 60 seconds. There are no specific requirements on the flag, but it should be recognizable as a flag.

3. ACTION SEQUENCE LIST (ASL)

- a. Teams will use the template provided to create an Action Sequence List corresponding to their device.
[w ESO Mission Possible Action Sequence List 26-27.docx](#)
- b. Teams must submit TWO copies of the Action Sequence Lists (ASL) to the Event Supervisor during impound.
 - i. Your ASL must describe the sequence in which the Action Transfers will occur (this must include a brief description, not just a list)
 - ii. Your ASL must indicate the Form of Energy used in each action
 - iii. Your ASL must have your team number clearly marked on it.
- c. The ASL may be kept by the supervisor for use in scoring and/or score counseling.
- d. Link to ASL template. Be sure to save a copy so that you can edit the document.
[w ESO Mission Possible Action Sequence List 26-27.docx](#)
- e. Teams without an ASL will be placed in Tier 2.

THE COMPETITION:

1. IMPOUND:

- a. All devices must be impounded during the indicated impound time. Teams who do not impound may not be allowed to compete or will be placed in Tier 2 based on the supervisor's discretion.
- b. Adults may be able to help with impound but will not be able to assist during testing blocks. Teams should ensure that students are able to move their device from the impound location to the testing location. If adults enter the testing area, the team may be placed in Tier 3.
- c. Each device must pass a safety inspection BEFORE operation. Uncontrolled or hazardous non-shielded, falling off launch objects, hazardous materials or spills, hazardous flammable substances, faulty wiring, or any other potential hazard can result in being unable to compete and earning only participation points for the event (Tier 3)
- d. Two copies of the ASL will be impounded with the device.
- e. Teams should impound goggles for use during testing.
- f. The team must also impound any materials which may be needed during the testing process or to repair the device. No other materials will be allowed.

2. COMPETITION:

- a. The event will be held indoors.

- b. Teams will have a maximum of 5 minutes to move and reset their device for testing. Teams will earn a bonus of 20 points if they start their device within 2 minutes of their competition time.
- c. Within 5 minutes of their competition time, the team will notify the supervisor that they are ready to test. Students must have their goggles on to begin the run. Teams who do not have goggles will be given one warning and allowed to get goggles but their 5 minute time will not be stopped. Teams without goggles will be placed in Tier 2.
- d. The team must follow the directions of the supervisor to start the device.
- e. Teams should not touch their device unless the device stops, jams, or fails. In that case, the participants are allowed to touch/adjust it to continue operation. Participants do not need to wait for the event supervisor's permission but should notify the supervisor of the restart.
 - i. Teams will lose 10 points for each restart needed during the run with a maximum of 3 restarts allowed before time is called on the run.
 - ii. Obvious stalling or adjusting only to impact runtime will result in disqualification (Tier 3).
- f. Teams will be allowed to run their device only one time.
- g. Run time ends when:
 - i. The final task is complete
 - ii. The device has been restarted 3 times and stops (Time will be recorded as 2 minutes and only successful scorable actions will be awarded points)
 - iii. Two minutes have passed
- h. The event supervisor may interview students about their device. Therefore, team members should be prepared to identify scorable actions on their device and identify the forms of energy and energy transformation taking place.

RELATED FLORIDA STANDARDS:

- **SC.5.P.10.1** Investigate and describe basic forms of energy, including light, heat, sound, electrical, chemical, and mechanical
- **SC.5.P.10.2** Investigate and explain that energy has the ability to cause motion or create change
- **SC.5.P.10.4** Investigate and explain that electrical energy can be transformed into heat, light, and sound energy, as well as energy of motion.

SCORING :

1. **High score wins**
2. The Action Score will be calculated using the following:
 - a. 40 points for a starting action
 - b. 20 points for each scorable action
 - c. 40 points for final action
3. The runtime score will be 60 points minus 1 point for each second below or beyond the 60 second target time.
4. Bonus points will be awarded for early start as stated above.
5. Penalties will be deducted for restarts and violations for materials outside of device as stated above.
6. Ties will be broken by the quality/accuracy of the ASL.